
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

Anonymous Author(s)

Affiliation

Address

email

Abstract

Generative image models trained on large-scale datasets now produce striking images, yet they still fail on seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that should plausibly be present in their training data. Because these failures are usually reported anecdotally, it remains unclear whether they are isolated quirks or evidence of broader, measurable limitations. We make this gap explicit by introducing a systematic framework for identifying and characterizing “conceptual blindspots”: concepts that appear in natural data but are suppressed or exaggerated in model generations. Our method uses sparse autoencoders (SAEs) to extract interpretable concept embeddings and compare concept prevalence quantitatively between real and generated images. As a methodological contribution, we train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—so that fine-grained conceptual disparities can be measured at scale. As an empirical contribution, applying the framework to four popular generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky) reveals suppressed blindspots such as bird feeders, DVD discs, and whitespaces on documents, as well as exaggerated blindspots such as wood background texture and palm trees. The same representation also supports datapoint-level analysis, where we isolate memorization artifacts: cases in which models reproduce highly specific visual templates seen during training. Together, these contributions provide a theoretically grounded and reviewer-auditable way to assess conceptual fidelity with respect to the underlying data-generating process.

1 Introduction

Generative image models trained on large scale datasets have become a core visual technology, supporting applications within computer vision [1–9] and increasingly beyond it [9–15]. This progress makes their persistent concept-level failures especially important: qualitative [16–18] and quantitative studies [19, 20] show that models still struggle with concepts such as human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted with these concepts, models often produce related visual structure without realizing the intended concept itself—for example, producing hands with six fingers. We refer to these failures as “conceptual blindspots”¹—cases where a model should plausibly have observed accurate demonstrations of a concept, yet systematically omits or distorts it at generation time. The central question for this paper is therefore not whether a few memorable failures exist, but whether concept omission and exaggeration form a systematic phenomenon that can be measured across a broad concept space.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ is our concept-level diagnostic: it measures whether a model over- or under-generates concept c_k relative to its natural-data frequency. We label concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The examples, generated by four popular generative image models, preview the paper’s main empirical contribution: blindspots and aligned concepts can be surfaced and compared at scale.

Existing evaluation tools do not make this contribution structure possible because they rarely expose failures at the concept level [28, 29]. Coarse-grained measures such as FID [30] primarily assess image realism and therefore obscure which concepts are missing or overrepresented. Metrics such as CLIPScore and distribution-coverage statistics [31–33] provide useful but partial evidence, since they do not isolate fine-grained features or concepts [34, 35]. Qualitative evaluations, including participant surveys [5, 17, 36–38] and open-ended exploration [39], can reveal individual failures but do not yield a scalable, repeatable basis for comparing model families.

This work. We contribute a framework that makes conceptual blindspots measurable. The key requirement is an automated and unsupervised procedure that identifies concepts whose likelihood under the true data-generating process differs from their likelihood under a trained generative model. We organize the paper around three reviewer-facing contributions:

- **Formalization.** We define suppressed and exaggerated conceptual blindspots by comparing concept prevalence in generated images against natural images from the data-generating process (Section 3). This turns anecdotal failures into a principled quantity that can be estimated and compared.
- **Method.** We instantiate the definition with sparse autoencoders (SAEs), extracting comparable concept distributions from natural and synthesized images (Section 4). As part of this contribution, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Tool and findings.** We build an exploratory web tool for distribution- and datapoint-level blindspot analysis across models (Sections 5.1–5.4). Applying it to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5) surfaces suppressed blindspots such as bird feeders, DVD discs, and whitespaces on documents, and exaggerated blindspots such as wood background texture and palm trees, as previewed in Fig. 1.

The model, code, and web tool are available at a software repository, supporting inspection and reuse of the contribution beyond the experiments reported here.

2 Related Work

2.1 Explainability in Vision

Our work builds on explainability methods that move from pixel-level attribution toward concept-level auditing. Early explainable-AI work in computer vision attributed predictions to influential input regions [40–42], but such methods offered limited semantic information and could produce incorrect explanations [43–45]. Concept-based interpretability [46] addressed this limitation by identifying semantically meaningful directions in neural networks, revealing not only where a model looks but also what concepts and structures it uses [47–49].

This paper specifically contributes a scalable SAE-based audit. Recent work shows that concept-based methods such as ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51] can be viewed as related dictionary-learning objectives under different constraints [52]. Among them, SAEs are especially attractive for large concept vocabularies. Although original SAEs can learn overly specific features [53], struggle with compositionality [54], and have limited intervention effects [55], newer

75 variants such as archetypal SAE (RA-SAE) [56] improve the stability and interpretability needed for
76 our contribution.

77 Other interpretability tools, including prompt-based probing [57], attention-map or activation visu-
78 alizations [58, 59], and dataset-level statistics [32, 33], offer useful but incomplete support for our
79 goal. Their main limitation for blindspot discovery is granularity: they typically operate over full
80 images or prompts rather than fine-grained features and concepts [34, 35]. They also often require
81 subjective interpretation and do not clearly distinguish among different kinds of failure modes [60].
82 These gaps motivate our contribution: a repeatable procedure for systematically identifying feature-
83 and concept-level weaknesses of generative image models [28].

84 2.2 Generative Image Models

85 We evaluate modern generative image models because diffusion-based methods now dominate
86 generative vision across images [4, 61-64], video [65-72], and 3D [7, 73-75]. In image generation,
87 this line traces back to denoising diffusion probabilistic models (DDPMs) [76], later extended to
88 non-Markov diffusion processes through denoising diffusion implicit models (DDIMs) [77].

89 The Stable Diffusion (SD) [9] family made DDPM-style image generation widely accessible to
90 research and open-source communities. Subsequent versions include SD 2 [78], SD 3 [79], and
91 SD XL [80], alongside extensions that constrain style [81, 82], pose [83], identity [84, 85], or input
92 modality. Related latent diffusion models, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88],
93 and FLUX [89], make cross-model blindspot comparison an important empirical contribution.

94 2.3 Datasets for Generative Image Models

95 Large-scale internet data is central to both the success of generative vision models and the blindspots
96 we study [90-92]. LAION-5B [93], scraped from Common Crawl [94], has been especially influential
97 for open-source text-to-image models such as SD and its derivatives; it contains over 5B image-
98 caption pairs, including 2.3B in English. Other major resources include COYO-700M [95] and
99 Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

100 The same datasets also introduce evaluation challenges that motivate concept-level auditing. As
101 LAION-5B became widely used, concerns grew about bias and harmful content [97-101]; filtering
102 did not remove all problematic material [97-99], including NSFW content [100] and hundreds
103 of CSAM instances [101], prompting temporary removal from official channels. LAION-5B also
104 contains low-quality images [102] and internet-style captions, such as product descriptions, that can
105 diverge from how users prompt trained models [103].

106 2.4 Concept Discovery and Sparse Coding in Generative Image Models

107 Our methodological contribution relies on dictionary learning: sparse representations in which each
108 input is reconstructed from a small number of dictionary atoms [104-106]. Building on compressed
109 sensing theory [107, 108], the field progressed from vector quantization [109] to Non-negative Matrix
110 Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent work adds online
111 methods [114], structured sparsity [115], theoretical guarantees [116, 117], and connections to deep
112 learning [118, 119].

113 Sparse coding has already been used to study high-level concepts inside diffusion models [120].
114 Earlier concept-grounded approaches for generative architectures include concept-bottleneck models,
115 which require human intervention at training time [121], and post-hoc detectors that retrofit concept
116 supervision [122]. Their shared limitation is that they require human-defined concepts, and therefore
117 miss broader blindspot patterns that a user has not already named.

118 Aggregate metrics such as precision, recall, density, and coverage [35, 123], as well as latent density
119 scores that predict sample quality from the model’s latent space [124], provide another route to
120 evaluating generative image models. These metrics can reveal distributional gaps, but they do not
121 identify which specific concepts are under- or over-represented.

122 Together, this prior work motivates our contribution: a scalable, unsupervised framework that
123 systematically identifies and quantifies concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

Our first contribution is a formal definition of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and model-generated outputs. The definition is designed to support a quantitative measure of which concepts are under- or over-represented by a model relative to the data distribution. Fig. 2 illustrates the formal setup. The construction uses standard assumptions [126–130]; empirically, we still observe meaningful blindspot phenomena when these assumptions are only approximate.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ for both the latent density $p(c)$ and its pushforward to image space $p(x)$, where $x = G(c)$. This convention keeps the blindspot definition compact and is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. In this formalization, individual latent dimensions represent the **Concepts** underlying the data distribution $P_{\mathcal{X}}$ over the image observation space $\mathcal{X}$. For example, different latents may encode concepts such as color, shape, size, location, and related factors [131–132]. We also allow the data-generating process to associate a text description $t \in \mathcal{T}$ with any sampled image, while leaving the text process itself implicit. These descriptions define the text-conditioned setting we audit: a **Generative image model** g_θ , with parameters θ , is trained on image-text pairs to map a noise signal $\eta \sim \mathcal{N}(0, I)$ and a scene description t to an image x illustrating that description.

The blindspot definition compares, concept by concept, the probability mass assigned by the data-generating process with the mass induced by the model g_θ . Operationally, we start from an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and generate a matched synthetic set $\mathcal{D}'_{\mathcal{X}}$ from the same text descriptions. To estimate concept occurrence, we use an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to concept energies, with the k^{th} coordinate representing the energy of concept k , i.e., $\xi_k(x) = E(c_k)$. This estimate is well-defined under the invertibility assumption on the data-generating process. This yields the probability that the data-generating process assigns to image x , written $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ with partition function Z_ξ , and the corresponding population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. We then define the paper’s central diagnostic, the average energy difference for the k^{th} concept between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$, using the sigmoid map $\sigma(\cdot)$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{x'}[\xi_k(x')] - \mathbb{E}_x[\xi_k(x)]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{x \in \mathcal{D}} \xi_k(x))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

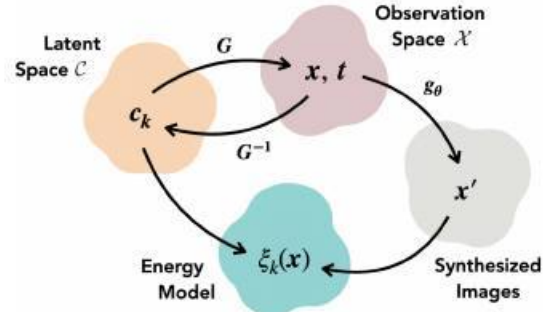


Figure 2: **Formal blindspot setup.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP. S_θ generates images x' based on t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , providing the common basis used to identify suppressed and exaggerated blindspots.

Thus, the energy difference for concept k summarizes how often that concept appears in generated observations ($\mathcal{D}'_{\mathcal{X}}$) relative to the natural baseline ($\mathcal{D}_{\mathcal{X}}$). This diagnostic directly supports the formal definition of conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). *Given a generative image model g_{θ} , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_{\theta} \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_{\theta} \leftrightarrow G}(k) > \lambda_{\max}$.*

In short, a conceptual blindspot is a concept whose likelihood in generated images deviates sharply from its prevalence under the data-generating process, either by suppression or exaggeration. Suppressed concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout the experiments, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these two regimes.

This definition is related to “mode collapse” in prior work (e.g., see Bau et al. [39]), but its contribution is a finer granularity of analysis. Mode collapse concerns exaggerated or suppressed odds of generating *entire images*; conceptual blindspots instead measure changed odds for specific concepts. For example, if a model fails to produce an object with a `white` background, the missing concept is a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

Our second contribution is the pipeline that operationalizes the definition for a generative model g_{θ} . Following Sec. 3, the pipeline requires two objects: (i) generated images from g_{θ} that can be matched to natural images from the data-generating process, and (ii) an energy model that places both sets in a shared concept space. For clarity, we use $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_0$ for the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix X , $X \geq 0$ denotes element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix A by A_i .

From Prompts to Latent Representations. The first stage creates the paired evidence needed for blindspot measurement in a model g_{θ} . We compare natural image-caption pairs (x, t) from $\mathcal{D}_{\mathcal{X}}$, sampled from the data-generating process G , with synthetic counterparts produced by the generative model g_{θ} from the same text descriptions.

Given t , we synthesize a counterpart image x' with a pretrained text-to-image generator $g_{\theta} : \mathcal{T} \rightarrow \mathcal{X}$, implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling follows the standard latent-space reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, I), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, I),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded by a pretrained VAE to yield $x' = \text{VAE}(\gamma_0)$. For the rest of the paper, this sampling detail is abstracted away and g_{θ} is treated as a black-box prompt-to-image map, $t \mapsto x'$.

Defining the Energy Model. The second stage contributes a scalable proxy for the energy model in Sec. 3. Motivated by work showing that self-supervised representations can invert the data-generating process and identify its underlying energy up to linear transformations [126, 128, 133, 134], we use DINOv2 [135] as the visual feature extractor. Because the number of concepts underlying the DGP may exceed the dimensionality of the feature space [51, 136], we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces corresponding to individual concepts [50, 56, 137, 139]. The intuition is that if DINOv2 represents concepts through approximately orthogonal directions, matching the independence constraint in Def. 1, then an SAE can isolate those concepts along individual latent dimensions [136]. The activation of an SAE dimension then serves as our approximation to the ground-truth energy of the concept represented by that dimension.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. We compute features $a = f(x) \in \mathbb{R}^d$ for natural and synthetic images in $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$. With n image-text pairs, we stack real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. The SAE contribution is to decompose each feature vector into a sparse combination of learned concept atoms. Let $D \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse

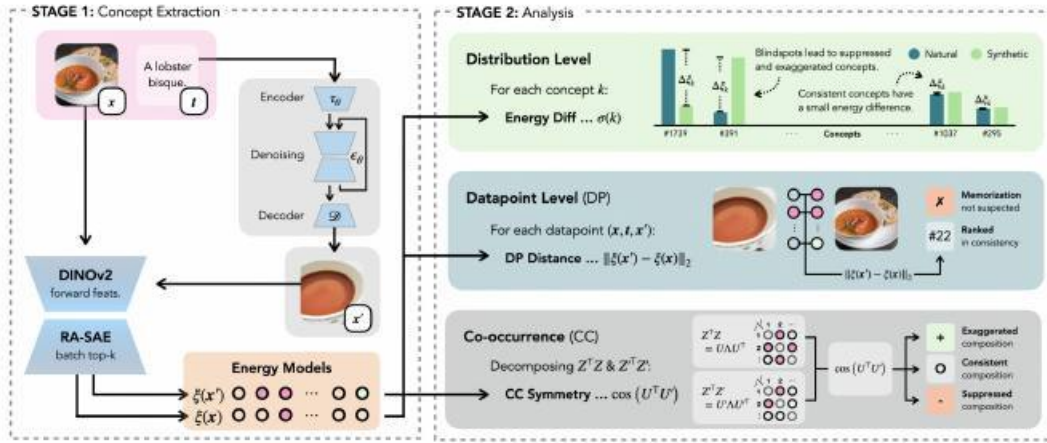


Figure 3: **Concept extraction contribution.** For a triplet (x, t, x') , we map x and its generated counterpart through DINOv2 and then through RA-SAE to obtain sparse concept embeddings. These embeddings instantiate the energy models $\xi(x)$ and $\xi(x')$ used throughout the results; in particular, $\xi_k(x) = E(c_k)$ stores the energy of the k -th concept. Applying Ψ row-wise gives the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K}$, where each row $z_i = \Psi(a_i)$ is the concept decomposition of one image. The decoder reconstructs features as $ZD^\top$, and the SAE is trained to minimize reconstruction error under sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

For blindspot analysis, dictionary stability is essential: vanilla SAEs can drift toward arbitrarily oriented dictionaries, making downstream conclusions sensitive to random seed. To make the study reproducible and seed-robust, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE contributes this stability by constraining the dictionary D to be a convex combination of training data. Specifically, we write $D = WA$ with $W \in \Omega_{K',n}$, where $\Omega_{K',n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K',n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W1 = 1\}. \quad (3)$$

Thus every atom D_i lies in the data convex hull $\text{conv}(A)$, and every reconstruction $ZD^\top$ lies in the data conic hull $\text{cone}(A)$. This constraint keeps learned concepts faithful to the support of the data distribution [56], which is crucial when the resulting codes are interpreted as concept energies. After training, the SAE provides a consistent sparse coding basis: Z for real images and Z' for their generated counterparts. *These codes express the same prompt-conditioned visual semantics in a shared, interpretable concept basis, and each activation value serves as a concept energy for blindspot analysis.* In summary, the method takes a prompt and associated real image (t, x) and returns paired sparse concept vectors (z, z') , enabling direct comparison between natural and generated content in a common conceptual basis.

This pipeline is the bridge from the formal contribution to the empirical contribution: starting from a captioned image (t, x) , we synthesize x' , map both images into a shared sparse concept space, and use the interpretable representations (z, z') to evaluate conceptual shifts induced by generation.

5 Results

Our empirical contribution evaluates four LAION-5B-trained generative image models—SD 1.5, SD 2.1, PixArt, and Kandinsky—on $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). The results are organized around three levels of evidence (Fig. 3): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces ambiguity, omission, and memorization; and a **compositional** analysis uncovers distortions in concept co-occurrence geometry.

To make these findings inspectable, we contribute an interactive exploratory tool, shown



Figure 4: **Evaluation pairs used for blindspot analysis.** Representative LAION-5B image-caption pairs are shown with matching generations from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

in Fig. 5. The web interface centers on a UMAP projection of concept representations and supports visualization and comparison of concept-level energy differences. It is publicly available at a software repository together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses below are derived from the following tool capabilities:

- **Contrast models and architectures.** For each evaluated model, the tool shows a UMAP visualization over all 32,000 RA-SAE concepts, with each point representing one concept and color indicating its energy difference.
- **Inspect individual concepts.** Each concept card reports key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Rank blindspots.** Beyond the UMAP and per-concept views, global rankings expose the most salient suppressed and exaggerated conceptual blindspots.

Rather than exhaustively studying a single failure mode, we use the results section to demonstrate the tool’s range and to surface the main blindspot patterns it enables others to inspect.

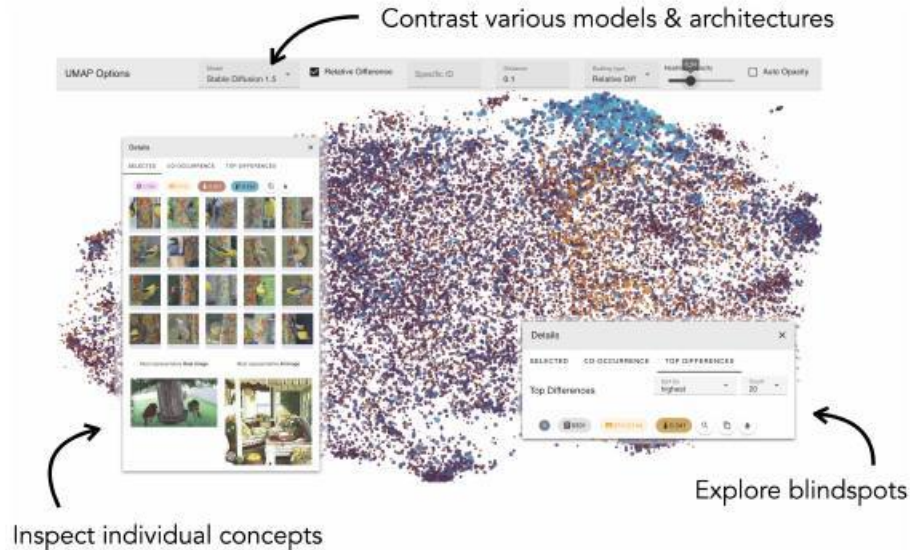


Figure 5: **Exploratory tool contribution.** The web interface displays a UMAP projection for each evaluated model; each dot is a concept, color-coded by energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, representative natural and generated images x and x' , co-occurrence rankings, and global lists of suppressed and exaggerated blindspots.

5.1 Generative Image Models Cannot Produce Unseen Concepts

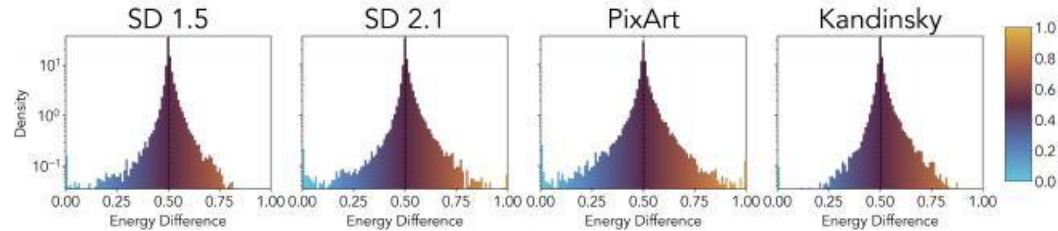


Figure 6: **Distribution-level blindspot evidence.** Log-scale histograms of energy differences $\delta(k)$ across 32,000 concepts compare natural and synthesized distributions for each evaluated model. Values left of zero indicate suppressed concepts; values right of zero indicate exaggerated concepts.

We first quantify the distribution-level contribution by analyzing the marginal energy difference $\delta(k)$ across the 32,000 RA-SAE concepts. As defined in Sec. 4, this quantity reports the relative prevalence of each concept in synthesized images compared with natural images. A value of $\delta(k) < 0.1$ marks concept k as under-represented, or suppressed, while $\delta(k) > 0.9$ marks it as over-represented, or exaggerated. Fig. 6 presents the $\delta(k)$ distribution for each evaluated model. Across all models, the histograms are heavy-tailed and place substantial mass near both extremes, providing direct evidence of systematic concept-coverage discrepancies. The left tail, corresponding to suppressed concepts, is denser and longer than the right tail, indicating that omission is a particularly salient failure mode. This asymmetry is reflected by negative distribution skewness: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

The shared qualitative pattern does not imply identical blindspot profiles. For instance, PixArt has a wider spread, suggesting stronger concept suppression. The consistent left-skew across models motivates the next contribution of the results: moving from marginal distributions to the structure, specificity, and examples of the omitted or exaggerated concepts.

5.2 Structure and Specificity of Conceptual Blindspots

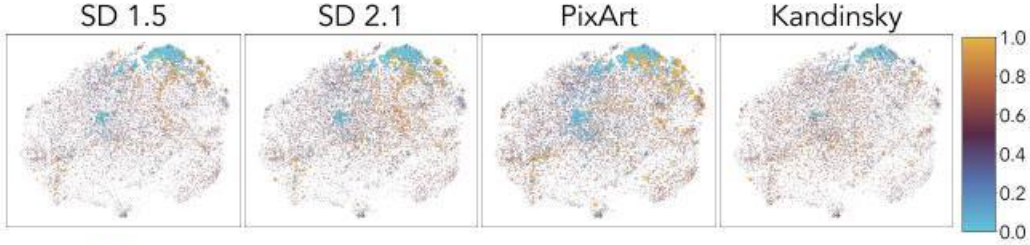


Figure 7: **Structured blindspot geometry.** UMAP visualizations of 32,000 concepts are colored by energy difference $\delta(k)$ between natural and synthesized distributions. Clusters reveal that blindspots are structured rather than isolated, with suppressed concepts on the blue end and exaggerated concepts on the yellow end.

After establishing that blindspots exist in aggregate, we next ask whether they have structure in concept space. We embed all 32,000 concepts into two dimensions using UMAP on the sparse codes and color each concept by its $\delta(\cdot)$ value. Fig. 7 shows distinct clusters across all models. These clusters often form contiguous regions of conceptual blindspots, especially for suppressed (blue) concepts, showing that the failures are structured rather than randomly scattered and may reflect shared training-distribution biases or architectural priors. To assess how much of this structure is shared across architectures, we compare both the magnitude and organization of concept-level $\delta(\cdot)$ values across models. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vectors of SD 1.5 and the other models. As expected, SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), consistent with their shared architectural and training pipelines. Their correlations with PixArt and Kandinsky are substantially lower— $r = 0.41$ and $r = 0.46$, respectively—indicating model-specific emphasis in different regions of conceptual space.

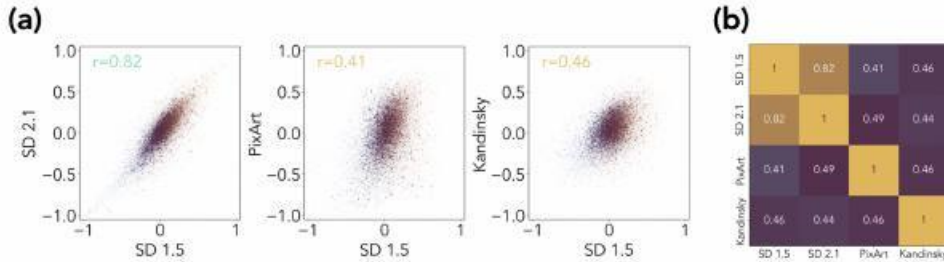


Figure 8: **Shared versus model-specific blindspots.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models report Pearson correlations in the top left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations for other architectures, indicating model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** The heatmap summarizes pairwise Pearson correlations for δ across models, quantifying how much blindspot structure is shared.

Together, these results sharpen the empirical contribution: some blindspots are shared across models, plausibly reflecting dataset-level effects, while others are model-specific and likely tied to training dynamics or capacity. The next section therefore turns these structured patterns into concrete examples of shared and unique blindspots.

5.3 Qualitative Examples of Blindspots

We next make the distribution-level contribution visually concrete by showing examples of suppressed and exaggerated blindspots. Fig. 9a shows a suppressed blindspot shared by all evaluated models: each model fails to reproduce the concept **solid white on documents**. Although the caption explicitly references the concept, none of the generations capture the intended visual semantics, indicating a systematically under-sampled region of concept space. Fig. 9b shows the complementary case of a model-specific blindspot: **pan** is captured by three models but is missing from Kandinsky generations. This example reinforces the structural analysis in Sec. 5.2: cross-model agreement can be high for some concepts and limited for others.

Conversely, Fig. 10 shows an exaggerated blindspot in which the concept **shadow under animal** is overemphasized in generated images. Although shadows are plausible in these scenes, their consistent and pronounced rendering across models, compared with the more nuanced occurrences in natural images, suggests an overactive prior. Despite searching for concepts exaggerated by only one model, we did not find clear examples, suggesting that salient exaggerations in this analysis are approximately universal.

These examples show the practical value of the diagnostic: it surfaces both shared and model-specific failure modes in concrete generated images. In particular, it reveals surprising limitations, such as the consistent failure to reproduce clear or solid background elements like whitespaces on documents, suggesting that some failures may arise from architectural constraints or training-data biases that transcend individual models.

The aggregate analysis also motivates a datapoint-level contribution: determining whether blindspots only appear statistically across many samples or whether they can be traced to individual prompt-image pairs. This finer-grained view helps distin-

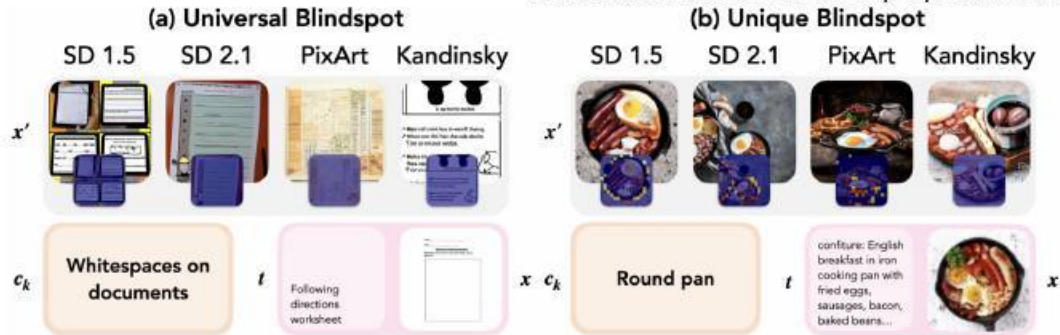


Figure 9: **Suppressed blindspot examples.** Natural images x , representative concept c_k , and caption t are shown alongside four synthesized images x' generated using S_θ . The universal blindspot appears in all evaluated models, whereas the unique blindspot appears only in Kandinsky.



Figure 10: **Exaggerated conceptual blindspot example.** Four synthesized images $D'_{\mathcal{X}}$, archetypal for the concept shadow under animal, are shown with the corresponding natural image $D_{\mathcal{X}}$ and caption t used for generation. The accentuated, nearly solid black shadows underneath animals illustrate a universal blindspot present in all

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misinterpreta-
memorization,
concept
omis-

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we inspect natural vs. generated image pairs with the largest and smallest average $\delta(\cdot)$ differences across concepts. This analysis makes successes and failures directly inspectable; failures often arise from prompt ambiguity or memorization artifacts. Fig. 11a shows instances with near-zero average $\delta(\cdot)$ differences. At the concept-energy level, these generations are nearly indistinguishable from the originals. Qualitative inspection shows that this alignment can reflect replication rather than faithful prompt understanding: repeated visual structures, such as clothing outlines or object arrangements, suggest copying from frequent training patterns. By contrast, Fig. 11b shows samples with some of the largest $\delta(\cdot)$ values. These examples indicate substantial conceptual divergence between synthesized and natural images. Some discrepancies arise from under-specified or noisy captions, but others reveal genuine blindspots: the prompt describes a clear concept present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. This failure suggests that, even with adequate language grounding, certain concepts remain outside the model’s effective generative abilities.



Figure 11: **• Datapoint-level conceptual alignment.** (a) Minimal energy-difference examples reveal cases where models appear to memorize training patterns. (b) Large energy-difference examples reveal concept divergences caused by prompt ambiguity or model limitations.

368 5.5 Analyzing Post-Training Effects

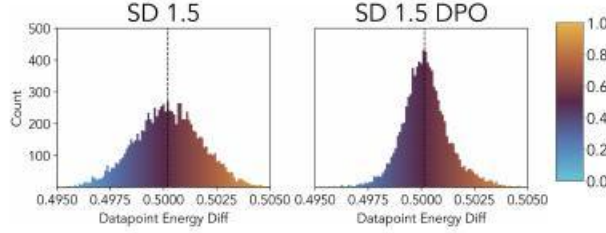


Figure 12: **Post-training effect on concept fidelity.** Histogram of datapoint-wise energy differences between synthesized and natural distributions for SD 1.5 models with and without DPO.

369 Our framework can also evaluate post-training interventions, an additional use case for the contri-
 370 bution. Post-training protocols such as safety fine-tuning have been argued to reduce generation
 371 diversity [140]. Because the pipeline isolates concept-level differences between a model’s generations
 372 and the ground-truth DGP, we use it to study DPO, a popular safety fine-tuning protocol [141].
 373 Specifically, we compare two SD 1.5 variants, one trained with DPO and one without. For each
 374 image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 distance between their internal concept-energy vectors,
 375 $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 shows the resulting histogram of datapoint-wise energy differences.
 376 The DPO-enhanced model has both a lower median and a narrower spread, indicating that generated
 377 concepts align more consistently with the ground-truth DGP. This suggests that DPO may regularize
 378 the model’s concept distribution, encouraging outputs that more closely reflect the semantic content
 379 of the inputs. Although this analysis does not disentangle the specific inductive biases introduced by
 380 DPO, the results provide empirical evidence that an objective favoring human-preferred generations
 381 can indirectly improve match to the training distribution. In particular, DPO reduces *both over and*
 382 *under* activation of individual concepts relative to the natural baseline. These findings show that the
 383 pipeline characterizes post-training interventions not only by output quality, but also by how they
 384 reshape the conceptual geometry of the model’s output space.

385 5.6 Conceptual Misalignment as a Function of Empirical Frequency

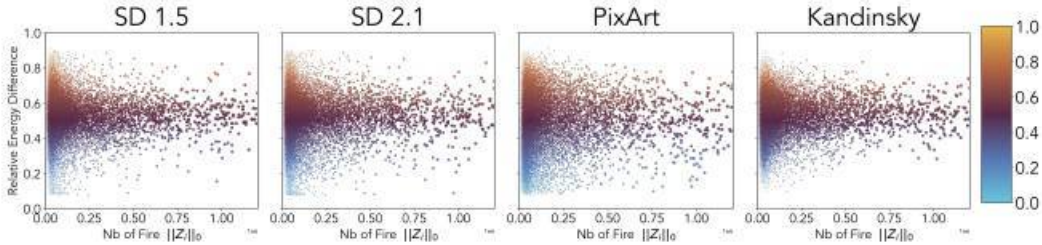


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots relate natural-data concept frequency (x-axis) to energy difference (y-axis) for the four evaluated models. Each point is a concept, with point size proportional to activation frequency.

386 We finally use the framework to probe a possible source of blindspots: they may not be only
 387 architectural artifacts, but may also arise from distributional peculiarities of certain features. We
 388 test this hypothesis by asking whether concepts that activate rarely in natural images, i.e., long-tail
 389 concepts, are more likely to become blindspots in generative models. Specifically, we process $\mathcal{D}_X$
 390 through the trained SAE and compute, for each concept k , its empirical frequency $\|Z_{:,k}\|_0$, where
 391 $Z_{:,k}$ is the activation vector for concept k across all images. We then compare this frequency with
 392 the absolute energy difference observed in generated outputs. Fig. 13 visualizes the relationship for
 393 all evaluated models. Concepts that occur more frequently in natural data tend to have lower energy
 394 discrepancies, whereas rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—show substantial
 395 alignment errors. This finding makes the results section more explanatory: many blindspots appear
 396 tied to long-tail concept distributions rather than random quirks, suggesting that mitigation may
 397 require data reweighting or augmentation in addition to architectural changes.

6 Discussion

This paper contributes a formal definition, an SAE-based measurement pipeline, and an exploratory tool for studying conceptual blindspots in generative image models. Applied to four popular models, the framework surfaces suppressed and exaggerated concepts, detects memorization artifacts, identifies datapoints with insufficient captions, quantifies post-training effects, and characterizes conceptual shifts across architectures. Each finding could support a dedicated investigation; here, we emphasize the breadth of analyses made possible by the framework and open space for follow-up work on deeper causal explanations, broader model coverage, and hierarchical RA-SAE representations for more nuanced concept analysis.

Limitations. The same design choices that make the contribution scalable also bound its scope. Because concept extraction relies on DINOv2 and RA-SAE, the analysis is limited to concepts that these models represent well; concepts poorly captured by this feature pipeline may remain invisible. In addition, although 10,000 images is substantial for the analyses reported here, it may still undersample rare concepts, concept co-occurrence, and other compositional statistics in the long tail.

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Appendix

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A Experimental Setup

This appendix makes the experimental contribution reproducible by detailing the setup used to analyze four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space supporting our blindspot comparisons is built from 10,000 image-text pairs sampled from LAION-5B [93], which serves as the natural-image domain. Because the original release is no longer available due to concerns about CSAM and other unsafe content, we use a substitute release with additional unsafe-content filtering, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure has three steps: (1) load the LAION dataset with the Hugging Face `datasets` library, (2) validate URL structure and resource availability using HTTP HEAD requests, and (3) sample with replacement until obtaining 10,000 valid examples. This yields D_G with accessible (x, t) tuples, which is important because many URLs from the original release are no longer available. Additional synthesized examples appear in Appendix H.

A.2 Synthesized Images

For each evaluated model, we generate a synthetic dataset D_{g_θ} that is in one-to-one correspondence with D_G , producing triplets (x, x', t) . Given each natural pair (x, t) from D_G , we use the same text t to synthesize a counterpart image x' with model g_θ .

The synthesis process follows the standard text-to-image pipeline for each architecture, implemented with Hugging Face `diffusers`. Models are loaded at mixed precision (fp16), and all synthetic images are generated at 512×512 resolution with default parameters.

Stable Diffusion 1.5. We use the checkpoint at <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5>, a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>. Inference uses 50 steps and guidance scale 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant used in Sec. 5.5 follows the baseline SD 1.5 implementation but replaces the UNet with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint at <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference uses 50 steps and guidance scale 7.5.

Kandinsky. We use the checkpoint at <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference uses 100 steps and guidance scale 4.0.

PixArt. We use the checkpoint at <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference uses 50 steps and guidance scale 7.5.

A.3 Distribution Level Analysis

Section 5.1 To support the distribution-level contribution, we compute $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. Normalization applies a sigmoid transformation with temperature $T = 0.8$, and the resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 To make blindspot structure visible, we embed all 32,000 concepts into two dimensions using UMAP on sparse concept codes. Each point is one concept, colored by energy difference $\delta(\cdot)$, so suppressed and exaggerated regions can be scanned directly. Cross-model consistency is quantified with pairwise Pearson correlations between $\delta(\cdot)$ vectors, reported as scatter plots and correlation matrices; this reveals whether blindspots cluster in concept space and whether they are model-specific or universal.

886 **Section 5.3** To surface qualitative examples, we rank all 32,000 concepts by $\delta(\cdot)$ and manually
 887 inspect the extrema, including both suppressed and exaggerated blindspots. Suppressed examples
 888 satisfy $\delta(\cdot) < 0.1$, and exaggerated examples are selected from the high- $\delta(\cdot)$ regime. The displayed
 889 concepts are manually annotated by inspecting their most activating images and spatial attention
 890 patterns; Appendix D discusses efforts to automate this interpretation step.

891 **Section 5.5** To evaluate post-training effects, we compare SD 1.5 with and without DPO. For
 892 each image pair (x, x') , we compute the L2 norm between concept-energy vectors, $\|\xi(x') - \xi(x)\|_2$.
 893 Element-wise differences are sigmoid-transformed with temperature $T = 0.8$ before averaging,
 894 yielding datapoint-wise energy differences that quantify concept-space deviation from the natural
 895 counterpart. We visualize these differences as overlapping histograms for the two model variants.

896 **Section 5.6** For each concept c_k , we compute its empirical frequency $\|Z_{:,i}\|_0$, the count of non-
 897 zero activations across the natural dataset. We then apply sigmoid normalization with temperature
 898 $T = 0.4$ to the energy differences $\delta(\cdot)$. The resulting scatter plots place empirical concept frequency
 899 on the x-axis and sigmoid-transformed energy difference on the y-axis; point size is proportional to
 900 activation frequency and color is proportional to energy-difference magnitude.

901 A.4 • Datapoint Level Analysis

902 **Section 5.4** For each image pair (x, x') , we compute the L2 norm between concept-energy vectors,
 903 $\|\xi(x') - \xi(x)\|_2$. This scalar ranks datapoints by conceptual divergence: minimal divergence high-
 904 lights potential memorization artifacts, while maximal divergence highlights substantial conceptual
 905 failures. The ranking makes datapoint-level failure modes inspectable.

906 A.5 • Co-occurrence Analysis

907 **Appendix F** For natural and synthesized data D_G and D_{g_θ} , we analyze concept co-occurrence
 908 through the co-activation matrix $Z^T Z$, which stores pairwise correlations in concept usage. Spectral
 909 analysis via eigendecomposition exposes dominant conceptual directions. We assess alignment
 910 between natural and synthetic co-occurrence structures with cosine-similarity heatmaps over the
 911 top-100 eigenvectors of each matrix. In these 100×100 heatmaps, diagonal alignment indicates
 912 shared principal concept axes, whereas off-diagonal structure reveals rotations and mismatches in
 913 compositional geometry.

B Custom RA-SAE Training Details

This section documents the RA-SAE training details that support the method contribution. The custom archetypal SAE (RA-SAE) [56] is open-sourced at a software repository.

Dataset. We train the auto-encoder on the full ImageNet-1k training split, containing approximately ≈ 1.28 M RGB images. Each image is converted into 261 visual tokens with DINOv2 [135], and tokens are passed to the SAE without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary contains 32,000 concept dimensions, providing the concept vocabulary used throughout the paper. Sparse activations use top- k masking with $k = 5$, setting all but the five largest activations per input to 0. We initialize weights with Xavier/Glorot and train at mixed precision (fp16), with the final ten epochs performed at full precision.

Optimizer and Schedule. The model is trained for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . We apply linear warm-up for the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. Training uses MSE loss plus an auxiliary penalty for activations that never enter the top- k set, with $\lambda = 10^{-5}$.

929 C Computational Resources

930 This section reports the GPU resources required to reproduce the training and experiments. In total,
931 the paper used approximately 202 GPU-hours on NVIDIA H100s and H200s.

932 **RA-SAE.** RA-SAE training required approximately 24 GPU-hours on three NVIDIA H100s.

933 **Synthesized Images.** Generating the full $D_{g\theta}$ (Appendix A) took roughly 5 hours per generator
934 when distributed across four NVIDIA H200 GPUs. Across five generators, this totaled approximately
935 100 GPU-hours on NVIDIA H200s.

936 **● Distribution-Level Analysis.** Extracting distribution-level energy differences took about 3 hours
937 per generator on one NVIDIA H200 GPU, for a total of ~ 15 GPU-hours.

938 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
939 datapoint, and ranking datapoints per concept took approximately 3 hours per generator on one H200
940 GPU, totaling ~ 15 GPU-hours.

941 D Autointerpretability

942 To make concept labels more scalable, we explored whether Vision-Language Models (VLMs) can
 943 automatically describe concepts identified by RA-SAE. We queried ChatGPT 4o through the OpenAI
 944 API with several prompt variants and image formats for an image x'' in which a target concept
 945 was salient, then recorded the generated descriptions. For this qualitative exploration, we allowed
 946 the VLM to provide full reasoning; a deployment-oriented version could request only the final
 947 description.

948 First, we gave the VLM two inputs: the raw image x'' and a version of x'' with a heatmap superimposed
 949 to localize concept intensity:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DiNOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

950 In this example, the VLM did not interpret the heatmap correctly. Rather than describing the wood
 951 texture in the background, which is the actual salient region, it described the foreground ring, which
 952 received no RA-SAE attention.

953 To test whether the failure was due to prompt underspecification, we repeated the scenario with an
 954 expanded prompt explaining the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DiNOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

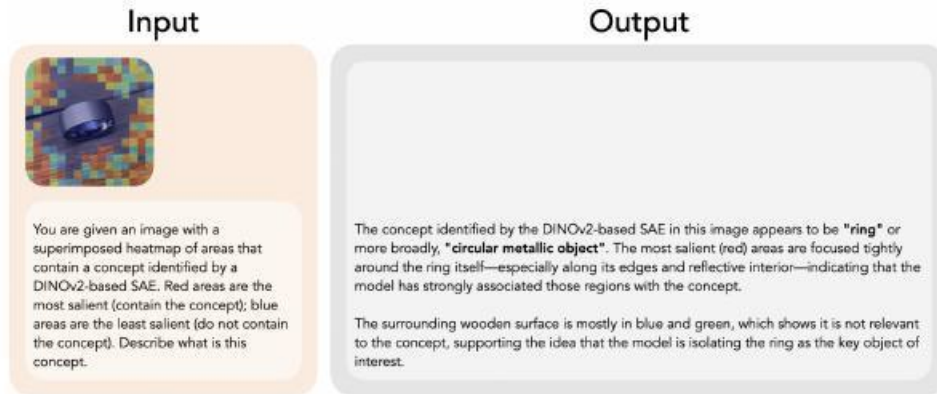
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

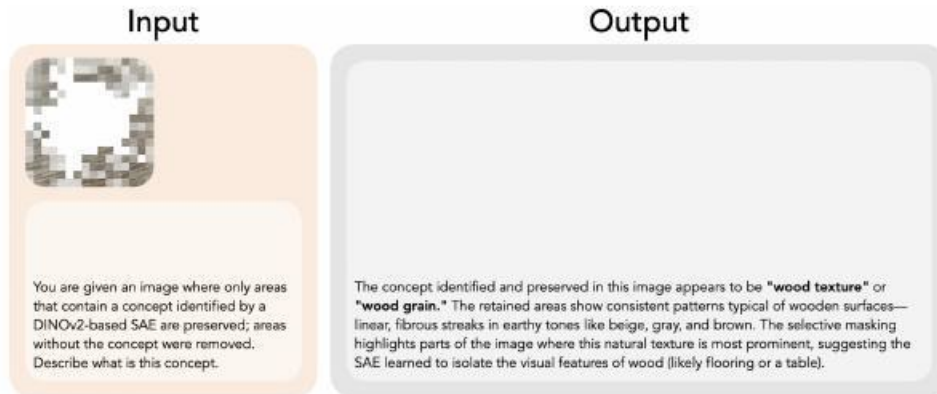
955 The expanded prompt did not resolve the issue: the VLM still identified the ring as the prominent
 956 concept. Additional prompting strategies, including more detailed heatmap descriptions and the code
 957 used to generate the heatmap, produced similar failures.

We then removed the raw image and provided only the heatmap-superimposed x'' with a similar prompt explaining how to interpret the heatmap:



This variant also failed to identify the wooden background and again focused on the ring. Further prompt variations with this image format did not materially change the result.

These observations suggest that the heatmap alone is insufficient to steer the VLM toward the salient region; irrelevant image regions can still dominate its description. We therefore replaced the heatmap-superimposed image with an alpha-masked version of x'' :



With alpha masking, the VLM correctly identified wood texture as the dominant concept, and we observed similar success across multiple concepts and images.

The image formatting and prompt shown above were the most reliable in our qualitative evaluation. This appendix therefore contributes a practical direction for scaling concept interpretation, but its evidence is limited by the absence of ground-truth annotations and by its focus on a single VLM. Future work on SAE autointerpretability can build on this analysis with broader models and quantitative validation.

972 E Additional Results: Model-Specific Blindspots

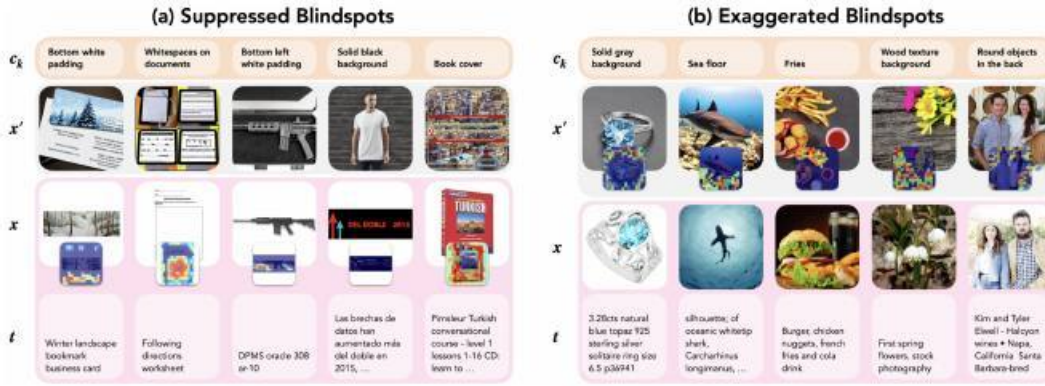


Figure 14: Model-specific blindspot examples for **Stable Diffusion 1.5**. For each concept, we show the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The superimposed spatial heatmap indicates where the concept is expressed.

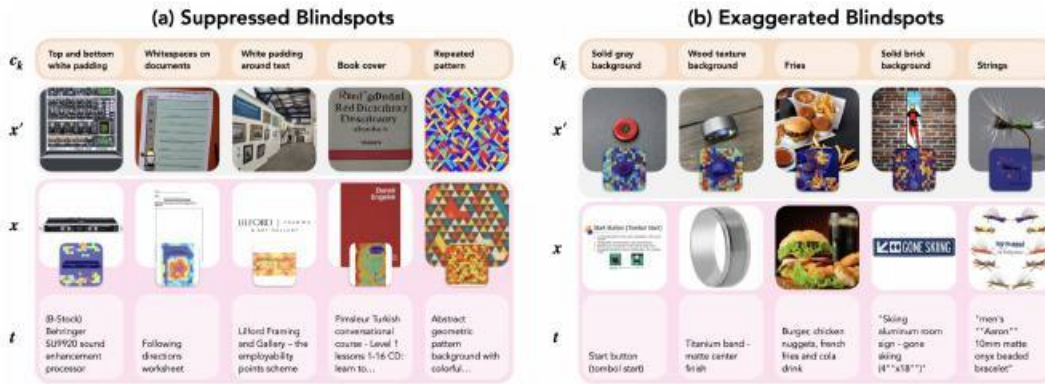


Figure 15: Model-specific blindspot examples for **Stable Diffusion 2.1**. For each concept, we show the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The superimposed spatial heatmap indicates where the concept is expressed.

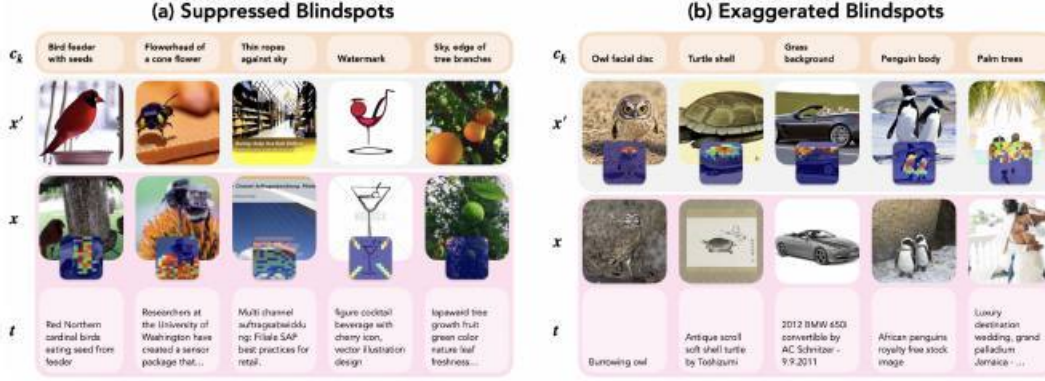


Figure 16: Model-specific blindspot examples for **Kandinsky**. For each concept, we show the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The superimposed spatial heatmap indicates where the concept is expressed.



Figure 17: Model-specific blindspot examples for **PixArt**. For each concept, we show the prototypical natural image for suppressed blindspots or synthesized image for exaggerated blindspots, selected by highest absolute activation. The superimposed spatial heatmap indicates where the concept is expressed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

The main results emphasize individual concept activations; this appendix extends the contribution to higher-order structure. Because visual scenes combine concepts through co-occurrence patterns that encode compositional semantics, we examine whether generative models preserve this structure using the co-activation matrix $Z^T Z$, which captures pairwise correlations in concept usage.

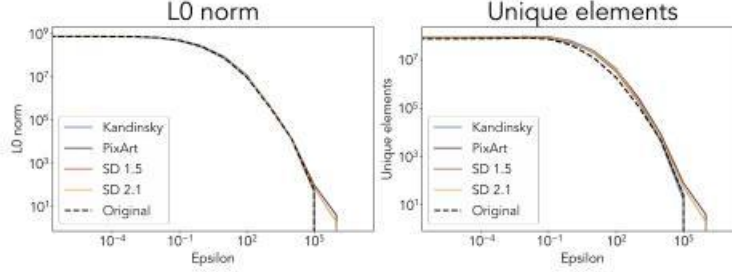


Figure 18: • **Sparsity and structural divergence.** Left: L0 norm of the co-occurrence matrix $Z Z^T$ as a function of ϵ (threshold), showing how many entries remain active in each model. Right: number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure but diverge in activation content.

At the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$, thresholded at varying ϵ values, tracks closely between natural and synthesized distributions across all models. This indicates that the coarse connectivity of the conceptual graph—which concepts co-activate at all—is well preserved. Formally, $Z^T Z$ can be viewed as the adjacency matrix of a weighted, undirected graph over concepts, with edge weights reflecting co-activation strength across the dataset.

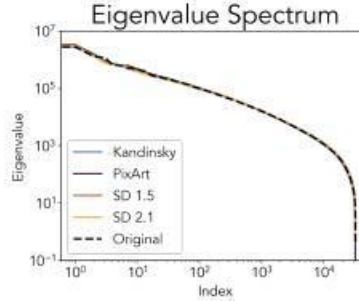


Figure 19: • **Spectral structure of co-occurrence.** Log-log plot of eigenvalue spectra from co-occurrence matrices $Z Z^T$ across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, Fig. 18 (right) shows that the content of these co-activations diverges: many entries in model-generated $Z^T Z$ are not shared with the natural baseline. Thus, models may preserve the capacity for compositionality while shifting which concept pairs become active, potentially reflecting model-specific inductive biases or training artifacts. To probe internal co-occurrence structure, we use spectral analysis. Fig. 19 shows eigenvalue spectra for each model and the natural distribution. All spectra exhibit heavy-tailed decay consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

We further examine dominant conceptual directions with cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). All models show partial diagonal alignment, implying overlap in principal concept axes, but off-diagonal entries reveal rotations and mismatches in higher modes that mark deviations in compositional geometry.

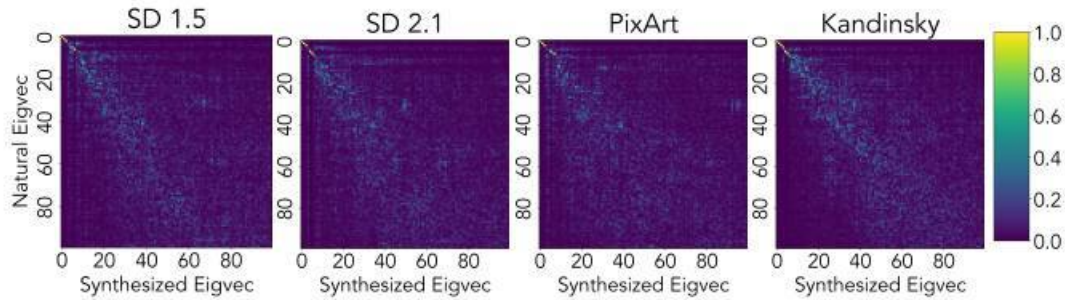


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps compare the top 100 eigenvectors of natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of dominant conceptual directions, with varying alignment across the four models.

997 Together, these appendix results extend the paper’s contribution beyond marginal blindspots: diffusion
 998 models approximate the global shape of concept co-activation, yet exhibit subtle and structured
 999 spectral deviations. Such higher-order discrepancies may underlie failures in coherent multi-object
 1000 scenes or relational concepts.

1001 G Additional Results: Blindspot Stress Testing

1002 To validate that identified blindspots correspond to observable generation failures, we stress-test
 1003 selected concepts with targeted prompts. We generate many images from prompts describing each
 1004 blindspot and compare models where the concept was identified as a blindspot against models where
 1005 it was not.

1006 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1007 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1008 *text-to-image model. Return these as a list of strings in Python.* We generated five images per prompt
 1009 and analyzed them with our custom RA-SAE model (Appendix B), ranking images by the intensity
 1010 of the desired concept. All images were then manually reviewed to determine whether the blindspot
 1011 was successfully depicted.

1012 The suppressed-concept examples below show partial but incomplete generation: models sometimes
 1013 produce a holder or string for the bird feeder blindspot, or a round hole for the glossy DVD
 1014 disc blindspot, but generally fail to produce the full concept. This behavior matches the method's
 1015 assessment and supports the stress-test validity.

1016 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Stress-test examples for prompts involving the bird feeder concept. Our method identifies this concept as a suppressed conceptual blindspot in Kandinsky, matching the observed behavior: Kandinsky fails to generate the corresponding concept. By contrast, SD 1.5, where this concept is not identified as a blindspot, generates it successfully.



Figure 22: Stress-test examples for prompts involving the glossy DVD disc concept. Our method identifies this concept as a suppressed conceptual blindspot in SD 1.5, matching the observed behavior: SD 1.5 fails to generate the corresponding concept. By contrast, Kandinsky, where this concept is not identified as a blindspot, generates it successfully.

1018 H Additional Examples: Synthesized Images



Figure 23: Additional evaluation examples: LAION-5B image-caption pairs with matching generations from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1019 I Concentration bounds for δ

1020 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1021 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1022 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid's
 1023 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1024 we can obtain fast and accurate estimates of concept bias.

1025 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1026 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1027 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1028 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1029 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1030 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1031 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1032 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1033 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1034 which proves the claim. □

1035 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1036 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1037 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by
 1038 no more than a small ε with high probability. This justifies our sampling strategy and confirms that
 1039 accurate bias measurements are attainable with limited data.

1040 J Monotonicity and Calibration-Free Interpretation of δ

1041 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In
 1042 practice, we use the score $\delta(k)$ for this purpose. However, one may wonder whether such a score
 1043 introduces distortions relative to more direct quantities such as the energy gap or the odds ratio.
 1044 The following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and
 1045 therefore inherits their ordering. This guarantees that no calibration is needed when using $\delta(k)$ to
 1046 rank concepts.

1047 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1048 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1049 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1050 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1051 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1052 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1053 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1054 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1055 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1056 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1057 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1058 concepts.

1059 □

1060 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1061 generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total
 1062 ordering over concepts.

K Stability of FID Under SAE Embeddings

This section supports the methodological contribution by showing how FID behaves after mapping activations through an SAE dictionary. We establish a quantitative relationship between FID in the original d -dimensional activation space and FID after applying a potentially overcomplete dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, D is assumed to have orthonormal columns but need not be square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin with the square, non-overcomplete case: if D is orthogonal and $k = d$, then the map from A to Z is an isometry and FID is exactly preserved. Because SAEs are typically overcomplete, we then handle the realistic case by bounding FID distortion using the extreme singular values of D and $D^\top D$. We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is $\mathcal{W}_2^2$ specialized to Gaussian surrogates.

For a probability measure μ on $\mathbb{R}^d$, write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically, we usually find $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on this observation. We first state the isometry lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This exact isometry is not the main regime of interest because SAEs typically rely on overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is not square orthonormal, but a column-orthonormal dictionary D still gives useful control: the next theorem bounds FID distortion through the extremal singular values.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1098 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1099 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1100 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1101 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1102 finishes the proof. $\square$

1103 Theorem 6 therefore clarifies the stability contribution of the SAE embedding: applying a column-
 1104 orthogonal overcomplete SAE dictionary cannot distort Fréchet Inception Distance by more than the
 1105 square of its extremal singular values. When $\mathbf{D}$ is nearly orthogonal—as we empirically usually find
 1106 with $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID measured in SAE feature space remains close to canonical FID. $\square$

1107